

15.3 Answers

① a) $\frac{9}{8}$

b) $\frac{1}{4}$

② a) $\frac{2}{3} [\ln 10 - \ln 3]$

b) $\frac{1}{2} e^4 - \frac{1}{2} e^{-\frac{3}{2}}$

c) $\frac{1}{12}$

d) $-\frac{1}{2} \cos 1 + \frac{1}{2}$

③ a) $\frac{7}{18}$

b) $\frac{171}{2}$

c) $\frac{16}{3}$

④ a) $\int_0^3 \int_0^{\sqrt[4]{3}} e^{x^2} dy dx = \frac{e^9 - 1}{6}$

b) $\int_0^1 \int_0^{\sqrt{y}} x^3 \sin y^3 dx dy = \frac{1}{12} (1 - \cos 1)$